

Book Reviews

The Bottom Line: Determining and Communicating the Value of the Special Library. Joseph R. Matthews. Westport, CT: Libraries Unlimited, 2002. 186 pp. \$42.00. (ISBN 1-59158-004-8).

Mr. Matthews knows his audience, and wisely consulted with Special Libraries Association (SLA) leaders, including former President Susan DiMattia and Former Research Committee Chair Sharyn Ladner (I was proud to serve on that Committee under Ladner's tenure). He also pays homage to the groundbreaking contributions to special libraries literature made by Joanne Marshall (also a past member of the SLA Research Committee) and Donald W. King with José Marie Griffiths. While Matthews, an established library automation consultant, is not well known in special libraries circles, these bona fides demonstrate his credibility and should put other special librarians' minds at ease as they did mine.

Each chapter of *The Bottom Line* begins with an appropriate epigram. The epigram used in the front matter succinctly sums up a key component of librarianship: "Smart people are not those that remember every detail of everything but those that know where to look for knowledge."—Tim McEachern and Bob O'Keefe."

On the first page of the introduction, Matthews articulates the challenge of proving value faced by most special librarians:

"Most librarians have personal, *anecdotal* [my emphasis] evidence of the experience of users who have benefited from having information readily and reliably available. The problem is that this evidence is uncertain at best and clearly *not quantifiable* [my emphasis]." (p. xiii).

Matthews further proves he knows special libraries when he cites key SLA evaluation studies (p. xv), and especially when he itemizes criteria non-librarian managers often want to see from library managers to demonstrate return on investment (ROI). These include: money; time; value-added information; prevention of litigation; timeliness and relevance of information; amount and accuracy of information; proactivity; consistent, error-free, and friendly service; and professional business image (pp. xv–xvi).

As Matthews comments, the criteria selected or preferred by the non-librarian manager may well not be "in tune" with (1) "the services of the library" or (2) "the information needs of the organization" (p. xvi). My staff and I are no exception. We have been increasingly challenged by the management team to whom I report to move forward company objectives that are not directly related to our core customers' ongoing information needs. Special librarians must carefully balance our responsiveness to both customer groups: the technical staff members who do the main work of the organization and the managers that lead the organization. Matthews' *The Bottom Line* can aid us in the balancing act, as he reviews and defines a variety of proven techniques for demonstrating the value of a special library.

Matthews favors two different evaluation techniques, the second broader than the first: Activity Based Costing (ABC) and The

Balanced Scorecard. In my library, we have used the former, and I have gained insights from Matthews on how to do so more effectively. I am not entirely sold on the detailed Balanced Scorecard approach but agree with Matthews that the goal-setting and measurability aspects of this approach are key for the best value analyses.

The eleven chapters of *The Bottom Line* are clearly organized and progress logically. Matthews establishes the characteristics of special libraries and of value measures, then discusses a variety of measurement techniques, culminating in two chapters focusing on the Balanced Scorecard approach. Throughout the text he makes good use of tabular displays and diagrams to illustrate measurements and outcomes.

In Chapter 1, Matthews lays out one of the key obstacles to non-librarian managers' understanding of the value of special libraries, though he does not identify it as such: the fact that information is a unique commodity that does not exhibit many of the properties of other economic goods (p. 3). Unlike most commodities, information cannot be divided, is not inherently scarce (one of the joys of our profession, to my mind!), frequency of use can increase rather than decrease its value, and information, while it may be expensive to produce initially, can be inexpensively be reproduced.

Matthews makes good use of Saracevic's and Kantor's work on applying the economic and philosophical concepts of value to libraries, research done for the Council on Library Resources and reported in this journal (June 1997), and later for SLA. As Matthews, Saracevic, and Kantor all note, "The benefits of the information to the user define its meaning" (p. 4). Thus, the study of the value of information can be broadly divided into three user-related categories, which Matthews concisely defines: normative value approach ("values of rigorous, formal" risk assessment models), realistic value approach (measures "the before and after consequences"), and perceived value approach ("assumes that users can recognize the value/[benefits] of information") (p. 405).

Matthews also relies upon the work of Robert S. Taylor in defining information users' requirements of information and, thus, measures of its value. These are: ease of use (see Moore's Law), noise reduction (communication theory-based), quality, adaptability, time-savings, and cost-savings (pp. 6–12).

In the succinct Chapter 2, Matthews defines various evaluation models and their associated measures, relying heavily upon Heron's work on quality, particularly his research for the American Library Association, and also on some of Donald King's evaluation work. He reinterprets King's evaluation model components Resources, Capability, Utilization, and Impact or Effect into broad types of measures:

- Input Measures
- Process measures
- Output measures, and
- Outcomes (pp. 16–19).

These types lay the groundwork for Chapters 3 through 8, in which he further defines techniques, then discusses each of the measures in detail, ending with a focus on outcomes in special libraries.

I found the definitions and guidelines in the substantive Chapter 3, Evaluation Techniques and Methods, especially useful. As Matthews notes, one needs to understand the reasons for evaluation (particularly if the evaluation was management's, and not your, idea!). Knowing the reasons can help you select the best methodology or methodologies. Specific methodologies defined include: surveys, sampling, interviews, observations, and case studies.

His list of desirable characteristics for measures is excellent and practical. A measure should be objective and unbiased; the data collection should be inexpensive; the measure should be appropriate for the service or activity; it should accurately reflect the characteristic being measured; it should be quantifiable; it must be able to be uniformly interpreted; and, it must be statistically reliable (pp. 22–23).

Matthews' definitions of evaluation measures are clear and useful, covering: counts/tallies, ratios (well-defined as "used to indicate the relative size of something"), formulas (The Clapper-Jordan formula for the Research Library Group is used to illustrate), questionnaires, weighted systems (e.g., the HAPLR Index for public libraries), list-checking, citation studies (e.g., ISI and others' impact analyses), content analysis, think-aloud protocol (mapping users' thought processes), and impressionistic (pp. 25–27).

In Chapter 3, he also reviews specific library services and ways to measure them. These include the collection, the online catalog (in and of itself and as a measure of collection use), reference services, technical services, and facilities. The remainder of the text considers overall evaluations of special libraries.

Input measures are familiar data in our profession, easier to measure than impacts and outcomes. In Chapter 4, Matthews categorizes input measures as including: budget, staffing, collection, automation system, and space (pp. 41–42). He has an excellent definition of standards in this chapter: "...the set of specifications or criteria used to measure the quality, shape, or other characteristic of a specific method or process" (p. 40).

Process measures like efficiency and staff productivity are also widely used in our profession, again being easier to capture than the more qualitative long-term measures. As Matthews says, "Process measures are concerned with what is done rather than what is achieved" (p. 45). These are discussed in Chapter 5, which also includes Matthews' delineation of Activity Based Costing (ABC), which if done correctly "...will reveal the links between performing a particular activity and the demand that activity makes on an organization's resources" (p. 46). He makes the good point that "[Library] products and services do not consume overhead resources equally" (p. 46). Some are extremely cost-efficient, as noted in the points about the reproducibility and simultaneity traits of information in the commodity discussion in Chapter 1.

Matthews favors ABC because "...the information provided by an ABC analysis forms the foundation for a number of performance mechanisms and systems and allows the library to have a better idea of how its inputs are being transformed into outputs" (p. 47). He discusses how ABC analyses can aid personnel and other cost reviews, as well as benchmarking studies and planned changes. I found his Scope of Change matrix (fig. 5.2, p. 52), illustrating the incremental to radical continuum applied to system changes, to be a useful tool I will apply in analyzing the effects of change in our company.

Output measures that describe the use of the library are covered in Chapter 6. These sorts of measures are also used fairly frequently in our profession. Matthews has changed my thinking on output measures, by his cogent statement in this chapter about their limitations:

Some have suggested that output measures have an outward focus, but in reality these measures reflect an *inward* [my emphasis] orientation in that they measure how much the library is used and *not the impact or effect of the library* [my emphasis] on the larger organization" (p. 56–57).

Yes! As he cautions us, "Remember, there is no clear inherent or implied value in an activity per se" (p. 57). Now my challenge, and that for all special librarians, is to develop the knowledge and skills to be able to conduct the more difficult, but more relevant, value analyses.

This key statement sets the stage for the remainder of the text, where Matthews discusses and demonstrates outcome and impact analyses (Chapters 7 and 8, as noted earlier), the Balanced Scorecard and its application (Chapters 9 and 10), and concludes with tips on communicating our value (Chapter 11).

Chapter 8 discusses the outcome or impact special libraries have on parent organizations in terms of accomplishments, time, and money. In the accomplishments section, Matthews draws from Joanne Marshall's groundbreaking 1993 special libraries impact study, especially in illustrating the avoiding-the-negative-consequences type of accomplishment (pp. 89–90). The Johns Hopkins incomplete literature review with tragic consequences cited is a case study that will be familiar to many special librarians.

In covering time aspects of impacts, Matthews notes the universal and ongoing value of special librarians in improving the productivity of their customers who are the organization's core staff—library staff are information experts (their customers are not, but they are subject experts) and have less expensive billable rates than their customers. Here he draws on Helen Manning's ROI study and on Griffiths' and King's studies to illustrate time and money saved by a library in several different ratios:

- Time/dollars saved against total budget, and,
- Typical library spending per employee and how much more it would cost for employees to do their own research.

Cost aspects of special libraries are divided into relative values and consequential values. In the relative values section, Matthews again draws on King's and Griffiths' work, discussing the derivation of values by service cost $\times$ number of times performed, then furthering that data with volume of activity data to estimate total value of a library. Consequential value data can be acquired by post-service surveys asking customers to estimate the financial impact of the service. Time as billable hours and/or direct monetary value can be queried.

In discussing the benefits a library provides to its parent organization, Matthews draws on several studies to indicate that freeing up customers' time can allow them to pursue more professional reading, an essential for advancement. He also discusses some of the medical information studies that identify qualitative benefits to library services. He makes a good point here that with a little analysis, qualitative benefits (e.g., reduced length of a patient's hospital stay as a result of a key information delivered to the physician by the library) can be shown to have quantitative, measurable aspects as well (e.g., lower costs to the patient and to some degree the hospital).

"One of the easiest traps for librarians to fall into is to rely on statements about intangible benefits and not think clearly about how the intangible benefits actually translate into an economic benefit" (p. 98).

Robert Kaplan and David Norton's Balanced Scorecard approach to valuing (first introduced in a 1992 HBR) article is defined and illustrated at length in Chapters 9 and 10. As noted by Matthews, Balanced Scorecard measures typically include four categories: financial, customer, internal business process, and innovation and learning. I like Matthews' explanation of the Scorecard, which has enabled me to see it more as a tool to integrate with other planning processes already in place, rather than as an additional, disconnected approach:

"The function of a balanced scorecard is to identify the key metrics across all levels of an organization with the fundamental purpose of aligning organizational goals and objectives with the activities at the operating levels" (p. 104).

Matthews' discussions of the four scorecard measures categories of financial perspective, customer perspective, internal process perspective, and learning perspective include excellent examples that special librarians can apply to each category—questions to get at measures, as well as measures themselves. He also notes how librarians can measure both the library's impact on the parent organization and changes within the library itself. For example, his question "Do staff have the proper tools, training and incentives to perform their tasks?" under the learning perspective (p. 113), can and should be used to identify gaps for both library customers and library staff. Then the library can proceed to acquire and develop resources to fill those gaps.

Matthews translates Frank Portugal's approach to valuing intellectual capital, from a recent SLA-supported study, into a tabular form that illustrates its similarity to the Balanced Scorecard (pp. 114–115). This is a good example of Matthews' direction to librarians to: "... present... important measures... in a cogent and understandable form for the management team of the larger organization" (p. 115).

In Chapter 10, Matthews presents useful tips on developing mission and values statements for special libraries, as well as strategic values and measures, and goals and action plans. The tips are presented as part of developing a library balanced scorecard, but are equally applicable to strategic planning processes that do not use the scorecard approach.

In Chapter 10, he also applies a PAPE (priority and performance evaluation) survey methodology to identify areas for improvement. And he covers ways to build new measures of progress towards enhancing these areas (pp. 131–132).

Chapter 11 on communicating the special library's value should be longer! More guidelines would be helpful to practitioners. Matthews does, however, provide some excellent tips, including: "educate don't bore," "focus on benefits," "use stories" to illustrate value, and use print materials ranging from bookmarks to annual reports to communicate ROI (pp. 144–146).

The Recommended Reading appendix of books and articles only cites one publication from the library literature. However, the references at the end of each chapter are primarily from the library field. Matthews' useful end matter also includes a glossary, input measure examples, and sample PAPE and benefits surveys.

The Bottom Line functions well as both a primer and a detailed guidebook to determine and demonstrate the ROI provided by a special library. Matthews concludes the text with an offset, boxed statement with which I strongly agree:

"The Bottom Line: Information obtained from special libraries is valuable" (p. 147).

Sara R. Tompson
Library Director
Packer Engineering, Inc.
1950 N. Washington St.
Naperville, IL 60563
E-mail: sarat57@msn.com

Virtual Inequality: Beyond the Digital Divide. Karen Mossberger, Caroline J. Tolbert, and Mary Stansbury. Washington, DC: Georgetown University Press, 2003. 192 pp. \$19.95. (0-87840-999-8).

Virtual Inequality is an important contribution to the digital divide debate. That debate takes two basic forms. One centers on the divide between the "information rich" developed countries and the "information poor" developing countries. The second is concerned with the rift between information "haves" and "have-nots" within countries. This book addresses the latter domain and is concerned with the digital divide in the United States.

This book is the product of a cross-disciplinary collaboration. Mossberger and Tolbert are both members of the Kent State University political science department while Stansbury is on the Library and Information Science faculty. The book is extremely well documented. Perhaps the chapter on the democracy divide and e-government is the best done, reflecting the political science bent of two of the authors. E-government is very well covered. Unfortunately, e-commerce and e-education go virtually unmentioned. If e-government is important to defining the digital divide, then certainly e-commerce and e-education are as well.

Mossberger, Tolbert, and Stansbury argue that the digital divide should be described as four different divides: the access divide, the skills divide, the economic opportunity divide, and the democratic divide. Each of these divides is developed in its own chapter. Each chapter draws well on the existing literature. The book is valuable if for no other reason than that it provides an excellent critique of the current state of the understanding of the digital divide in the United States. It is particularly good in its contrast of the approaches taken by the Clinton and George W. Bush administrations. Perhaps this is a function of the multidisciplinary strength of the book's authorship, for indeed it shows here.

The access divide is defined along "connectivity" lines: who has access to digital technologies. The authors confirm the conventional wisdom that age and education are important predictors of in-home access, but they also argue that race and ethnicity are also factors (pp. 32–33): Asian Americans have greatest access followed by whites, Latinos, and African Americans in that order. Most access the Internet from home or work, followed by friends' computers, libraries, and other access points.

The skills divide is defined as *technical competence and information literacy* (p. 38). Variation was found along technical competence for age, education, affluence, race, and ethnicity, but not gender (p. 47). The authors conclude that for the most part the skills divide mirrors the access divide (p. 55). While they found no gender difference, they did find a gender preference for skills acquisition: males prefer a more impersonal delivery ("online help and tutorials") while females prefer more personal instruction (p. 56).

The economic opportunity divide is predicated on the hypothesis that there has, indeed, been a major shift in opportunities driven by changes in the information environment. The authors document this paradigm shift well with arguments from the political and economic right and left. This chapter might be described as an "attitudinal" chapter. The authors are concerned here with the perceptions of their respondents of their information skills and skill levels with their economic outlook and opportunities. Technological skills and economic opportunities are correlated, one finds, in the minds of all across all ages, genders, races, ethnicities, and income levels. African Americans in particular are "... attuned to the use of technology for economic opportunity" (p. 80).

The fourth divide is the democratic divide. The Internet may increase political participation, the authors posit, but only among groups predisposed to participate and perhaps among those with the skills necessary to take advantage of the electronic environment (p. 86). Certainly the Web has played an important role in disseminating

and distributing political messages and in some cases in political fund raising. But by the analysis here, we must conclude that the message does not reach everyone equally. Thus, the Internet may widen the political participation gap rather than narrow it.

The book has one major, perhaps fatal, flaw: its methodology and statistical application. The book draws upon a survey performed for the authors in June and July 2001 by the Kent State University's Computer Assisted Telephone Interviewing (CATI) lab (pp. 7–9). CATI employed a survey protocol provided to the reader as Appendix 2. An examination of the questionnaire reveals that all questions yield either nominal or ordinal responses, including the income variable (pp. 9–10). Nevertheless, Mossberger, Tolbert, and Stansbury performed a series of multiple regression analyses (reported in a series of tables in Appendix 1) utilizing these data. Regression analysis requires interval/ratio data in order to be valid although nominal and ordinal data can be incorporated by building dichotomous dummy variables. Perhaps Mossberger, Tolbert, and Stansbury utilized dummy variables; but I do not find that discussed. Moreover, I would question a multiple regression made up completely of dichotomous dummy variables.

I come away from *Virtual Inequality* with mixed feelings. It is useful to think of the digital divide as more than one phenomenon.

The four divides that Mossberger, Tolbert, and Stansbury offer—access, skills, economic opportunity, and democratic—are useful as a point of departure and debate. No doubt, other divides will be identified and documented. This book will lead the way. Second, without question, Mossberger, Tolbert, and Stansbury provide us with an extremely well-documented, -written, and -argued work. Third, the authors are to be commended for the multidisciplinary of their work. Would that we could see more like it. My reservations about their methodological approach, however, hang over this review like a shroud.

Wallace Koehler

Director/Associate Professor

MLIS Program

Valdosta State University

Valdosta, GA 31698

E-mail: wkoehler@valdosta.edu

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